

SINGLE LINKAGE, COMPLETE LINKAGE AND K-MEANS CLUSTERINGS COMPARAISON ANALYSIS

Datasets: FOODP.DAT and MASST.DAT

DATA: FOODP.DAT

FOODP.DAT provides average prices of hamburgers, apples, bread, tomatoes, and butter in cents per pounds in 24 US Cities in 1978. The data is collected by the US Department of Labor, Bureau of Labor Statistics (Washington DC,1978). This data set is relevant because it shows the different average prices from five common food items within the same time frame, conditions, and quality. From an economic standpoint, high prices for these items would induce a higher consumer purchasing power. Based on customer purchasing power, the cost of living can be deduced in these metropolitan cities since the cost-of-living index is correlated to customer purchasing power.

Our clustering analysis would be unstandardized for this data sets because prices have the same unit (cents) and similar values within a reasonable margin of error. So, the original scale would remain the same. The goal is to cluster cities based on the singular effect on prices for these five variables. Initially, single linkage clustering would be useful for this dataset because we only have 24 cities and 5 variables. It should be noted that consumer purchasing power is a relative measure that needs to be analyze over time due to inflation. Therefore, a clustering analysis for the same products, recorded in the same conditions year after year would give a more detailed analysis of the evolution of cost of living and consumer purchasing power of clustered metropolitan US cities over time.

Procedure and Interpretation

- *Single Linkage Clustering*
- *Initial Step: Explore the data*

Process:

Cor function compute the correlation between vectors two vectors x,y in R. When both vectors are the same, the result is 1(identity). Therefore, the matrix diagonal is verified. City name is the categorical variable. It is set as row names as shown in the line of the code in **Table 1** for specific numerical values for each food item.

Interpretation:

It should note that hamburger and bread are highly correlated. Hamburger is also correlated with tomato. It can be deduced that some hamburgers have tomatoes, and all hamburgers have bread indeed (**Table 1**). On the other hand, both combinations apple & hamburger and apple & tomato have low correlations. Finally, the more complex the food item is, the more variance in average prices is shown. For instance, tomato has low variance while hamburger has high variance (**Table 2**). All this interpretation confirms our initial intuitions.

```

> rownames(data)=data[,1]
> data=data[,-1]
> ## Explore the data
> summary(data)
  Bread      Hamburger      Butter      Apples      Tomato
Min.   :28.90   Min.    : 84.5   Min.   :123.2   Min.   :35.60   Min.   : 75.90
1st Qu.:34.20   1st Qu.:107.0   1st Qu.:139.8   1st Qu.:46.42   1st Qu.: 84.17
Median :36.90   Median :109.8   Median :143.5   Median :51.10   Median : 89.90
Mean   :38.44   Mean   :112.2   Mean   :144.2   Mean   :51.74   Mean   : 89.76
3rd Qu.:40.20   3rd Qu.:117.1   3rd Qu.:150.5   3rd Qu.:58.08   3rd Qu.: 94.62
Max.   :70.90   Max.   :135.6   Max.   :162.3   Max.   :65.10   Max.   :104.50
> rho=cor(data)
> rho
      Bread Hamburger      Butter      Apples      Tomato
Bread    1.0000000 0.6490532 0.3301770 0.3187031 0.3620681
Hamburger 0.6490532 1.0000000 0.2447778 0.1908956 0.5557993
Butter    0.3301770 0.2447778 1.0000000 0.2351424 0.4361291
Apples    0.3187031 0.1908956 0.2351424 1.0000000 0.1333844
Tomato    0.3620681 0.5557993 0.4361291 0.1333844 1.0000000

```

Table 1: Summary Table and correlation matrix

```

> var1+var2+var3+var4+var5
[1] 415.0813
> percentvars
      Bread Hamburger      Butter      Apples      Tomato
0.1686394 0.3260335 0.2051058 0.1683354 0.1318860

```

Table 2: Variance for all five variables

- [*Step 1: Compute a distance matrix*](#)

Process:

The function `dist()` computes and returns the distance matrix computed by using the specified distance measure to compute the distances between the rows(city name) of our data matrix. The Euclidian method is applied here. The method is based on the Pythagorean theorem. It represents the shortest distance between two vectors (difference between the square root of the variable squared). The output is the matrix in the appendix (**Table 3**).

Interpretation:

We are looking at the distances between variables for any given observations. In other words, we are looking at the closeness for the 24 cities based on the average food prices. Anchorage and Honolulu have a low distance at 21. Both cities are isolated from mainland therefore they are “close” to each other. In the same way, Boston and Pittsburg have a low distance at 9 because these cities are on the East Coast. Food suppliers and supply chain cost are similar based on the region (**Table 3**). Since the variable are squared in the Euclidean method calculation, we must be careful with interpretation outliers (high distance vs low distance) as high squared value would dominate the other. This is indeed the weakness of this method which became relevant when we analyzed the cluster plot.

- [Step 2&3: Run hclust \(\) and plot the dendrogram](#)

Process:

`hclust()` function is described as hierarchical cluster analysis on a set of dissimilarities in R. Basically the algorithm calculate dissimilarities between new clusters and existing points based on the similarities prior to forming the new cluster(lance Williams dissimilarity update formula). Therefore, inversion is impossible while using single linkage method. The object value is defined by seven components i.e., merge, height, order and label, method, call and distance method that can be interpreted as follows. Height is $n-1$ real values with $n= 24$ (cities). And merge is defined by $(n-1) * matrix=46$. Order and labels are equal to n . class is defined as “none” for all components. And on step 4, assigned `hc` is plotted to get the cluster dendrogram.

```
hc<-hclust(D,"single")
summary(hc)

> summary(hc)
```

	Length	Class	Mode
merge	46	-none-	numeric
height	23	-none-	numeric
order	24	-none-	numeric
labels	24	-none-	character
method	1	-none-	character
call	3	-none-	call
dist.method	1	-none-	character

Table 4: `hclust` output. The data is in phase with the dimension of the data set. The full interpretation is provided by the dendrogram below.

Interpretation:

The lowest height is Cleveland and St Louis around height 6.5. Missouri and Ohio are Midwestern State. Agriculture is one of the biggest contributors of the economy of that region. Therefore, it makes sense that food prices are both cheap and similar between two cities in this region of the US. Matter of fact both states are part of the corn belt since 1850s (**Table 5**). At height 13.5, the lowest pairs are joined by 12 cities. Then all 14 cities join a pair of New York and Philadelphia. Both cities are on the East Coast. It fits our intuition that they have similar prices that are higher than the other 14. Finally, as expected, Anchorage and Honolulu were high on the far left. However, Kansas City and Buffalo were in between both cities which was unexpected. In conclusion Cities that paired up are in the same area of the country or highly connected on their transportation systems as it is the case for Dallas and Chicago in the latter case. If you compared to the standardized dendrogram (**Fig. 1**), some cities are still paired such as Cleveland Saint Louis and Chicago Dallas. However, a new pair such as San Francisco and Los Angeles while New York and Philadelphia are not a paired anymore. The standardized dendrogram was easier to read in my opinion. Even if it was unnecessary, standardizing the data is an appropriate approach as well.

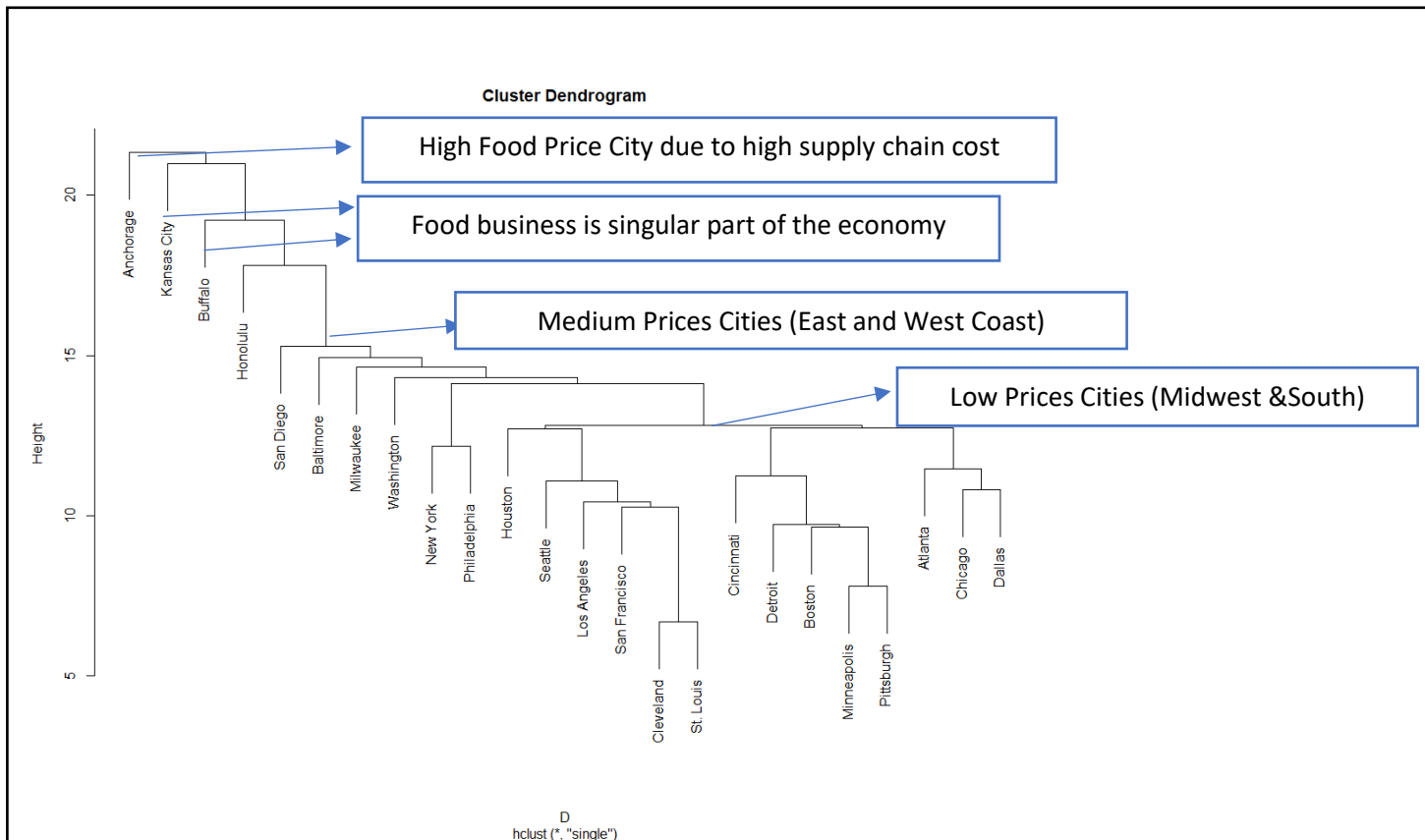


Figure 1: Cluster Dendrogram- It has been generated using hierarchical clustering. It should be noted that all 4 commentary boxes are intuitive approximations.

	Cleveland	St Louis	variation	% variation
bread	38.5	36.9	0.0415584	4.15584416
hamburger	107.7	109.8	-0.019499	-1.9498607
butter	142.7	140	0.0189208	1.89208129
apple	50.3	46.7	0.0715706	7.15705765
tomatoes	83.2	79	0.0504808	5.04807692

Table 5: Comparison of average prices between lowest pair of cities in dendrogram

Step 4: Choose number of clusters

Process:

The tree was produced by the function `hclust()` with seven components. `Cutree` return a vector with different groups based on k clusters. So, the tree is cut, and “close cities” are grouped together from $k=1$ to $k=n$. In this agglomerative technique, each city is considered as a cluster. Then, the algorithm adds pairs of data in succession. Therefore, two clusters that are the closest in Euclidean distance get together and created a new formed cluster. This process is looped until the termination condition is met for the 24 cities based on the food prices for the 5 variables.

Interpretation:

Indeed, clustered objects are along the x-axis while the distance at which the cluster is formed is on the y axis. East coast Cities that are clustered together such as Philadelphia and New York joins in cluster 2 with mainland cities. Honolulu was the latest city to join that cluster even though all food prices are higher than the ones in Kansas City and Buffalo. Both these cities had their own group with Anchorage which was expected. I could have chosen a higher k but I decide to keep it to 4 to show that most metropolitan cities joins the same cluster 2 in the mainland. However, Anchorage, Kansas City and Buffalo are different for the transportation, supply chain cost and food specificities of their cities. As it was the case for dendrograms, **Table 6** was the same for both for standardized and unstandardized variable due to the similar height in the y-axis in their respective trees (**Fig.2**).

```
> memb<-cutree(hc,k=4)
> memb
```

Anchorage	Atlanta	Baltimore	Boston	Buffalo	Chicago	Cincinnati	Cleveland
1	2	2	2	3	2	2	2
Dallas	Detroit	Honolulu	Houston	Kansas City	Los Angeles	Milwaukee	Minneapolis
2	2	2	2	4	2	2	2
New York	Philadelphia	Pittsburgh	St. Louis	San Diego	San Francisco	Seattle	Washington
2	2	2	2	2	2	2	2

```
> |
```

Table 6: Tables shows the four different groups for all 24 cities. Buffalo has the lowest price for tomato while Kansas City has the highest price for butter among all metropolitan cities. Those cities are known for their hamburgers, wings, barbecues, or food associated with our 5 variables.

Step 5&6: Get clusters centers and calculate sum of total SS. within SS for each cluster (compare to k-means below)

Interpretation:

Table 7 shows that only one entry at [1,] is positive for bread and this is similar for hamburgers. However, all the other entries are negative for these two variables. Once again, we see the clear correlation between bread and hamburgers in the cluster's centers determination. This is also true for the standardized dataset. Furthermore, TSS is much larger for single link than k-means for the dataset therefore the solution is worse because the ratio of BSS over TSS becomes small. Even though the dimension is small, single link is not better than k-means clustering for this data set. This interpretation is correct also if we used the standardized data (**Table 8**)

```
> cent<-NULL
> for (k in 1:4){cent<-rbind(cent,colMeans(data[memb==k,,drop=FALSE]))}
> cent # same as Table 7.21
```

	Bread	Hamburger	Butter	Apples	Tomato
[1,]	3.8795390	2.00755136	1.16909152	1.4550185	1.3977343
[2,]	-0.1432861	-0.03504964	-0.04882854	0.0746985	0.0345934
[3,]	-0.4711224	-0.20165057	-2.10394929	-1.9305539	-1.8730316
[4,]	-0.3994082	-1.06985833	1.96025702	-1.0931332	-0.2511642

Table 7: cluster centers (unstandardized)

- **k-means clustering**

Step 1: Run k-means and analyze clusters

Process:

The number of observations (n) is the number of metropolitan US cities. All observation is grouped into k clusters. In our case, the number has been determined to be 4 based on the dendrogram in the single linkage clustering analysis. In this k=4 clusters, each observation belongs to the cluster with the closest average. This cluster would be the prototype cluster computed in R.

Interpretation:

In Cluster 2, the means for all 5 variables is higher than all other clusters as it features Honolulu, Anchorage and two East Coast Cities i.e., Philadelphia and New York. Cluster 1 has the lowest prices in hamburgers and apples as several Midwest cities such as Cleveland and Saint Louis belong to it. Furthermore, cluster 4 has the lowest prices in bread and hamburgers while cluster 3 has balanced prices in comparison the other clusters.

To conclude clusters are different depending on the technique used. It is expected as both processes are unique. In other words, the analysis of all the clusters is clearer with k-means technique in my opinion. Furthermore, the quality of the partition is 58% which means that 42% represent the measure of total variance in the data set unexplained by clustering. Considering that TSS associated with k-means being lowered than TSS associated with single linkage. K-means clustering is confirmed as the most adequate technique (**Table 9 & Fig.3**).

K-means clustering with 4 clusters of sizes 7, 4, 9, 4

Cluster means:

	Bread	Hamburger	Butter	Apples	Tomato
1	35.41429	109.0286	134.5286	50.21429	83.62857
2	51.85000	131.2500	152.9750	57.10000	96.90000
3	36.41111	113.6556	145.4011	50.74444	94.13333
4	34.90000	95.7000	149.7250	51.27500	83.50000

Clustering vector:

	Anchorage	Atlanta	Baltimore	Boston	Buffalo	Chicago	Cincinnati	Cleveland
	2	3	3	3	1	3	3	1
	Dallas	Detroit	Honolulu	Houston	Kansas City	Los Angeles	Milwaukee	Minneapolis
	3	3	2	4	4	4	1	1
	New York	Philadelphia	Pittsburgh	St. Louis	San Diego	San Francisco	Seattle	Washington
	2	2	3	1	4	1	1	3

within cluster sum of squares by cluster:

[1] 1035.2200 880.3475 1441.4214 654.4750
(between_SS / total_SS = 58.0 %)

Available components:

[1]	"cluster"	"centers"	"totss"	"withinss"	"tot.withinss"	"betweenss"	"size"
[8]	"iter"	"ifault"					

Figure 3: k-means Output

Step 2&3: Plot clusters and choose k (plot total sum of squares)

Process:

Each cluster is assigned a color red, green, black, and blue assign to each 4 clusters. The quality of k means partition is calculated by dividing BSS-between sum of squares- per TSS-Total Sum of squares-

and multiplying by 100. Ideally, TSS should be much greater than BSS to get the best quality as the objective is to get internal cohesion and external separation.

Interpretation:

Cluster 2 members are represented by the red dots and the cluster 4 by the blue dots (**Fig.4**). When we look at the correlation between two variables, two red dots are consistently on the top right corner. It can deduce to be Anchorage and Honolulu. When we look at the variable butter, there is a blue dot consistently high therefore it can be deduced to be Kansas City. Furthermore, the correlation between apples & other variables and tomato & other variables are highly scattered. It depends on the fluctuation in demand for fruits and veggies.

Finally, using the elbow method on **Fig.5**, the optimal number of clusters is confirmed to be 4 as we plotted the explained variation in function of the number of clusters k. The elbow of the curve is the latest cluster point from which the modeling data fits. If we had any more cluster, we would get diminishing return in the model.

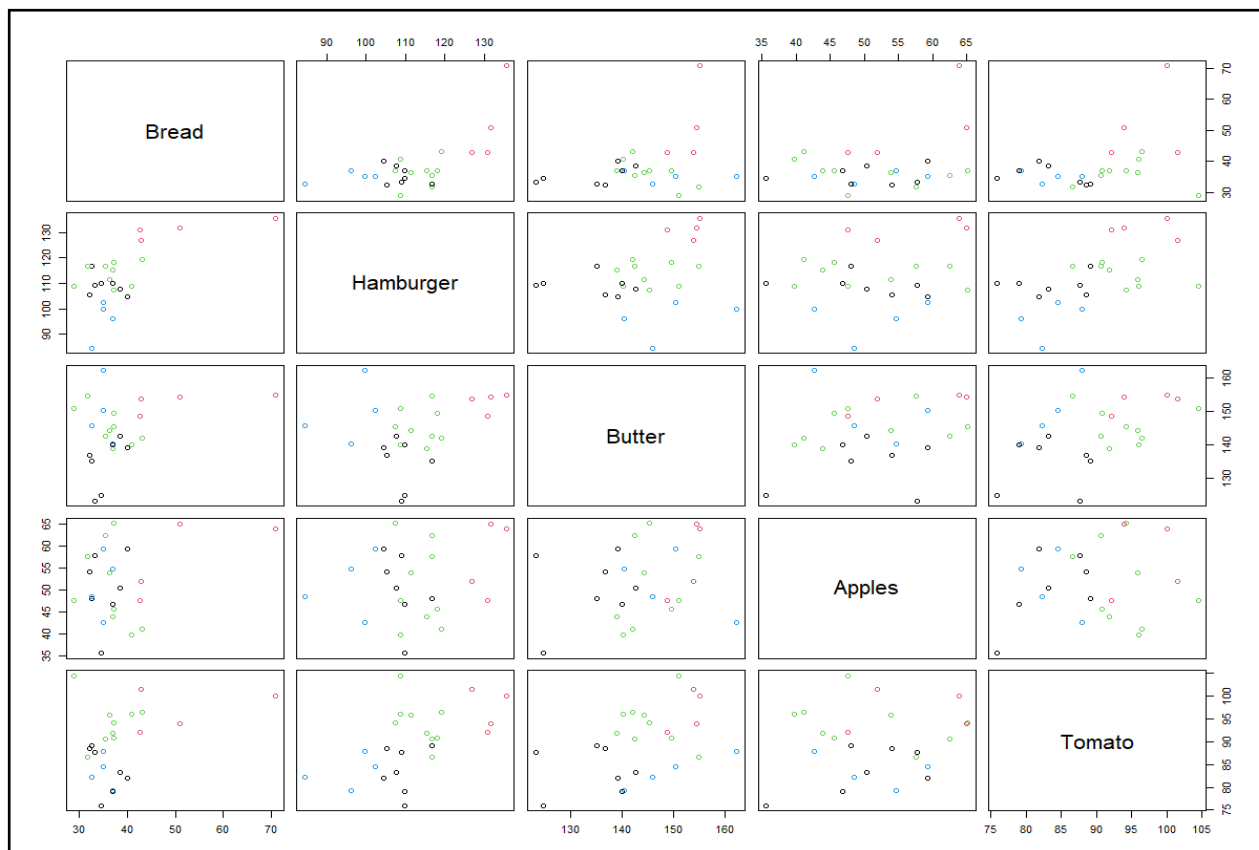


Figure 4: Cluster Plot(k-means)

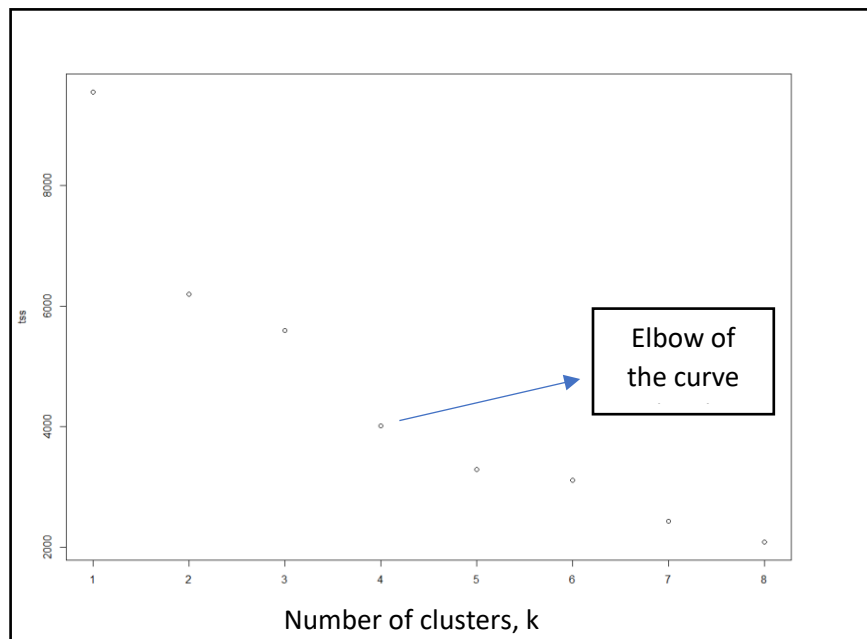


Figure 5: Total Sum of Square plot (TSS)

Step 4: Interpret clusters

- Interpretation:**

As discussed earlier, Anchorage and Honolulu are the driving forces of food prices for cluster 2 except for apples (**Table 10**). It can be deduced that the overall price of apples is lowered due to the Cities on the East Coast where apples are more readily available to customers in 1978. The demand is more important to analyze the price of the food items in New York and Philadelphia than the transportation cost. In my opinion, people would tend to eat more veggies and fruits then fatty foods in the East Coast in general. Furthermore, metropolitan cities inside the corn belt have low price for bread and hamburger (as some hamburgers are not 100% beef). Therefore, cluster 4 prices are driven by nearby farms therefore production cost are low. As shown is **Fig.6**, there is not any clear correlation between cities in cluster 3 and cluster 4 respectively. Indeed, the cities are located all over the mainland. Therefore, specific prices in cities skewed the overall means for these clusters.

```
> cl$centers
```

	Bread	Hamburger	Butter	Apples	Tomato
1	35.41429	109.0286	134.5286	50.21429	83.62857
2	51.85000	131.2500	152.9750	57.10000	96.90000
3	36.41111	113.6556	145.4011	50.74444	94.13333
4	34.90000	95.7000	149.7250	51.27500	83.50000

Table 10: Clusters Centers

```
> cl$cluster
```

	Anchorage	Atlanta	Baltimore	Boston	Buffalo	Chicago	Cincinnati	Cleveland
	2	3	3	3	1	3	3	1
	Dallas	Detroit	Honolulu	Houston	Kansas City	Los Angeles	Milwaukee	Minneapolis
	3	3	2	4	4	4	1	1
	New York	Philadelphia	Pittsburgh	St. Louis	San Diego	San Francisco	Seattle	washington
	2	2	3	1	4	1	1	3

Table 11: Cities grouped in 4 clusters(k-means)

Step 5: Plot clusters with PCA

Process: In a 3D plot, there are more principal components than PC1 and PC2 as shown in **Fig.7** By plotting one more component, R provides the component with the third most variation to optimize the analysis of the plot. Compared to FOODP.DAT, a 2D screenshot is useful due to the low dimension of the data($\dim(\text{data})=24$).

Interpretation:

Kansas City and Buffalo are skewing cluster 4 for different reasons (two outliers). Kansas City has the highest price for butter at 162.30 cents while Buffalo has the lowest price for apples and the second lowest for butter among all cities. As expected, Honolulu and Anchorage are located higher on the left of the plot due to food transportation costs. While being in the same cluster, these cities are relatively far away from Philadelphia and New York. Interestingly, the two clusters who have members from different area in the countries are more homogeneous than cluster 4 and cluster 2. It shows that extreme prices (too low or too high) dominate other cities in the clusters. This is explained by the formula of the Euclidean distance use as a default method of calculation of distance in R. Finally, it should be noted that cluster 1 and cluster 4 intersects in space (**Fig.6 & Table 11**).

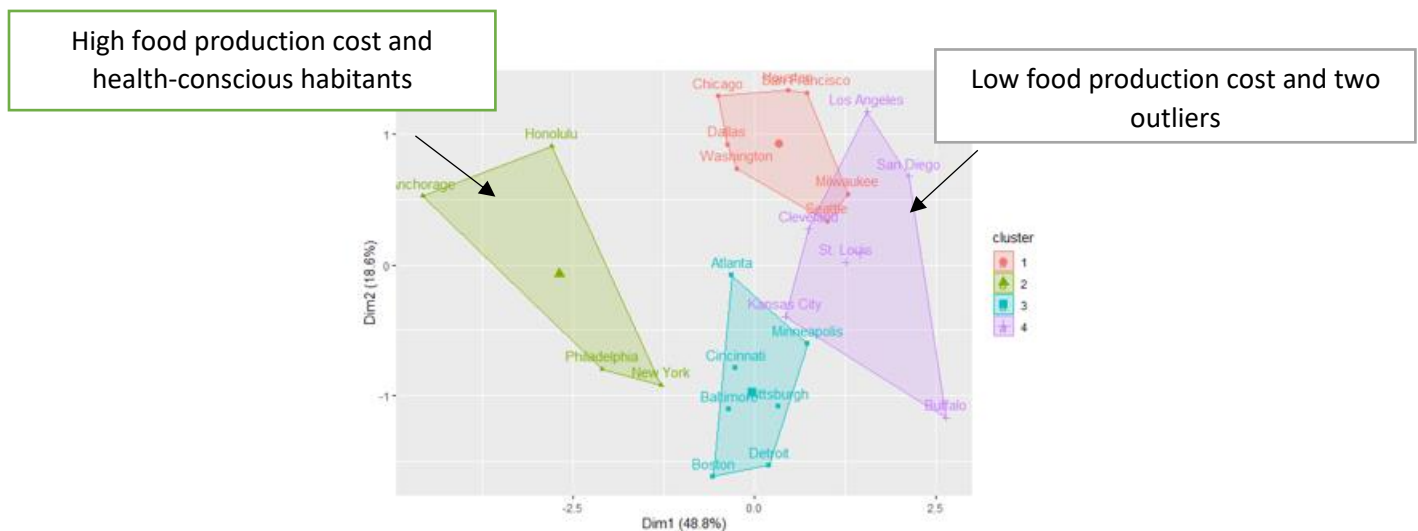


Figure 6: Annotated Cluster Plot

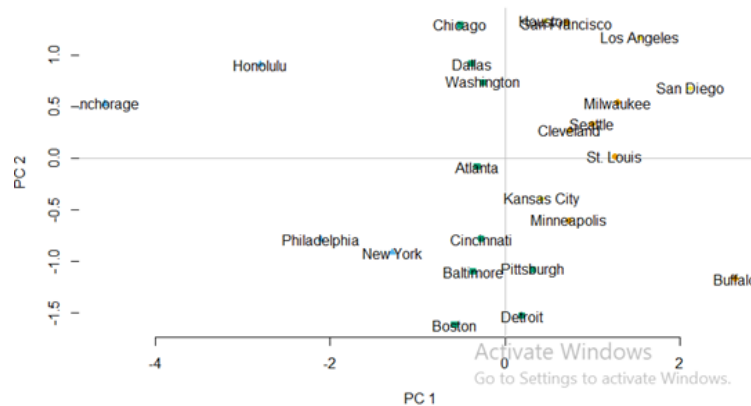


Figure 7: Plot Cluster 3D using k-means (2D screenshot). On this picture, anchorage is the further away than any other cities followed by Honolulu. Philadelphia and New York are close together while Buffalo is isolated on the right (higher PC1 value). As for Kansas City, it was distant from other cities on the z-axis.

DATA: MASST.DAT

MASST.DAT provides the result of a survey to estimate the potential demand for mass transportation. As oil prices are volatile, there was a sharp increase in oil prices during the Arab Oil Embargo in 1973-1974. Therefore, this survey was conducted in three southern cities to study the impact on this political decision on mass transportation demand. The survey is based on the demographics of the partakers and their views on mass transportation as oil prices have increased. The data set has 52 variables described in **Table 12**. There are two main differences between this data set and FOODP.DAT. First, there are over 1000 missing data therefore mean substitution would be used to analyze the data set. Furthermore, even though variables do not have different units, and all are following a numerical method of rankings; all variables should be on the same scale. Therefore, we are going to standardize the data. In other words, R would modify the variables to a mean of 0 and a standard deviation of 1. We want to compare the relevance of each variable with each other in the dataset. Single linkage clustering would be inappropriate due to the dimension of the data. This aspect would be shown on the last step while comparing both methods.

Procedure and Interpretation

- Single Linkage Clustering

Step 1: Compute a distance matrix

Process: As discussed above, we need to rescale the data. Each attribute is standardized by subtracting the mean value of that attribute and then dividing by the standard deviation to get the z-scores. In other words, we need to make sure that one variable does not dominate the other ones as you applied the Euclidean method. This would have been the case for variables V1 to V6 if the data has remained unstandardized. Additional operations are required as explained below.

Step 2&3: Run hclust () and Plot dendrogram

Process:

hclust function has been defined earlier as well as the processes to plot the dendrogram in R. As shown in **Table 13**, the responses are digits therefore you can apply mean responses to the missing data. It should be noted that mean substituted responses are estimates therefore there is a higher error associated with these values. The error change would be neglected in this process.

Interpretation:

After the initial step of standardizing the data, we must make some judgement calls to get the best model. First, we modified to a 1-to-5-point scale from low to high response by dividing each value by the adequate fraction. This was done for V39 by dividing it by 5 over 7, V40, V44 to V48 were divided by 6 over 5 and V52 was divided by 5 over 3. Finally, binary variables i.e., gender, marital status and driver licenses are not relevant for the model. After completing all these steps, we made the entire dataset homogenous (**Table 13**). Then we plotted the dendrogram. But since the dimension is high, the dendrogram is unreadable due to the single linkage method covered earlier (**Figure 8**). In the same way, the memb matrix also does not provide meaningful information as shown on **Table 14** below.

<pre>> hc<-hclust(D,"single")</pre>			
<pre>> summary(hc)</pre>			
	Length	Class	Mode
merge	1194	-none-	numeric
height	597	-none-	numeric
order	598	-none-	numeric
labels	598	-none-	character
method	1	-none-	character
call	3	-none-	call
dist.method	1	-none-	character

Table 13: hclust summary

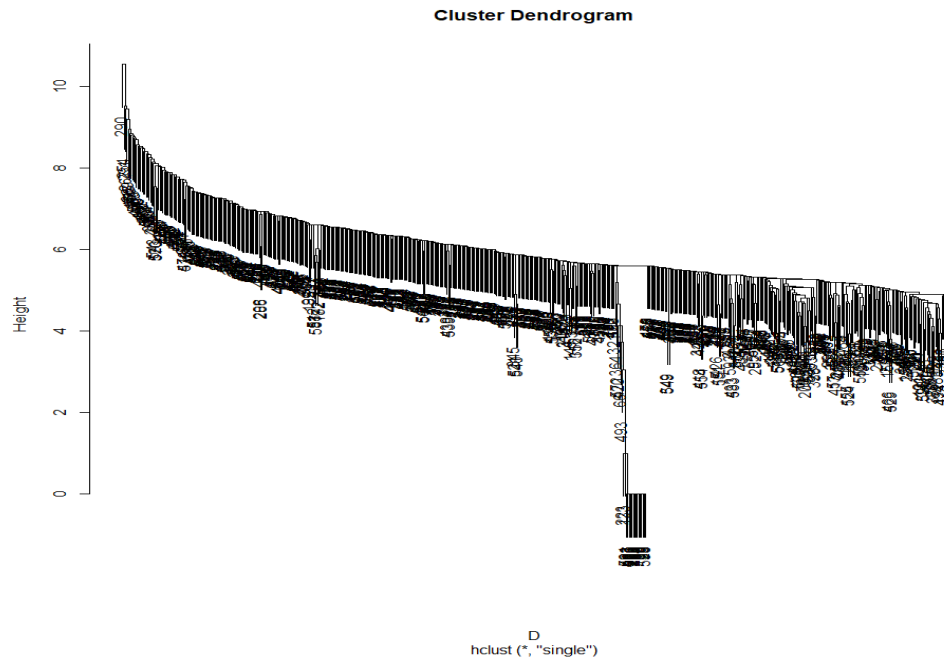


Figure 8: Cluster dendrogram. `dim(data)` is overwhelming for this dataset. There is not any insight in the dendrogram using single linkage clustering

Step 4&5&6: Choose number of clusters & get clusters centers and calculate sum of total SS. within SS for each cluster (compare to k-means below)

```

> membl <- ctree(hcc, k=2)
> membl
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66
2 2 2 2 1 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
2 2 2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165
1 2 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
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232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297
1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363
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364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396
2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
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2 1 1 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495
2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528
1 1 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594
2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
595 596 597 598
2 2 2 2 2
> centik=NULL
> for (k in 1:2){centik=rbind(cent1,colMeans(data[membl==k,,drop=FALSE]))}
> cent1
      V1      V2      V3      V4      V5      V6      V7      V8      V9      V10     V11
[1,] -0.26866676 -0.2392811 -0.1222288 -0.08754940 -0.4324528 -0.021975362 -0.6593792 -0.7910718 -0.9750192 -1.0792492 -1.1823506
[2,] 0.07858917 -0.06543989 -0.05495996 -0.02505117 0.1236508 0.006285409 0.1830080 0.2262635 0.2788765 0.361143 0.3381744
      V12      V13      V14      V15      V16      V17      V18      V19      V20      V21      V22
[1,] -1.2073140 -1.2168995 -1.1948411 -1.1845741 -1.1747696 -1.1173708 -1.07979325 -1.02312269 0.05978416 0.2581830 0.12447781
[2,] 0.3453178 0.3478295 0.3427914 0.3388137 0.3360094 0.3179789 0.3083119 0.0661006 -0.0145236 -0.0738459
```

long survey as such, the attention span of the participants would be focus on subject that affects their daily lives. For most people, there is a lack of knowledge of Middle East politics and oil prices fluctuations; but those variables are specific to the participants' bottom line. There are therefore more impactful than demographics questions or other opinions questions unrelated to their livelihood.

```
cl=kmeans(data,2) ## let's keep same number of clusters as before
> cl
K-means clustering with 2 clusters of sizes 386, 212

Cluster means:
      V1      V2      V3      V4      V5      V6      V7      V8      V9     V10     V11
1  0.08932341 -0.09531956 -0.03436879 -0.001504384  0.1620837 -0.02104906  0.1315031  0.2126471  0.3437548  0.4505381  0.5276737
2 -0.16263602  0.17355353  0.06257714  0.002739114 -0.2951146  0.03832517 -0.2394349 -0.3871782 -0.6258931 -0.8203194 -0.9607643

Cluster vectors:
      V1      V2      V3      V4      V5      V6      V7      V8      V9     V10     V11
1  0.5749467  0.0884232  0.6261832  0.6369651  0.6313387  0.6079724  0.589548  0.05960424 -0.06820888 -0.08172273  0.04088434
2 -1.0468370 -1.1077895 -1.1399803 -1.1597572 -1.1495130 -1.1069687 -1.073422 -0.10852470  0.12419165  0.14879705 -0.07298375

Cluster assignments:
      V1      V2      V3      V4      V5      V6      V7      V8      V9     V10     V11
1  0.03408286  0.003428466  0.05190012 -0.03390213  0.04615934  0.02939501 -0.03191736 -0.02851128 -0.03865900  0.009699424  0.009699424
2 -0.06205652 -0.006242395 -0.09449739  0.06172746 -0.08404484 -0.05333902  0.05811368  0.05191205  0.07038856 -0.017660272

Cluster centroids:
      V1      V2      V3      V4      V5      V6      V7      V8      V9     V10     V11
1  0.07383745 -0.01370287 -0.1092423 -0.07913491 -0.02591956  0.03721209  0.05190012  0.03181815  0.05635026  0.04040410 -0.03490126
2 -0.13441988  0.02494956  0.1989034  0.14408526  0.04719316 -0.06775409 -0.0946554 -0.05793304 -0.10260000 -0.07356596  0.06354663

Cluster variances:
      V1      V2      V3      V4      V5      V6      V7      V8      V9     V10     V11
1  0.01157188 -0.02188812 -0.04766410  0.0641089  0.03879437 -0.01688142 -0.003785975 -0.008828775
2  0.02100556  0.03985290  0.08678463 -0.1167266 -0.07063503  0.03673693  0.006893332  0.16075034

Clustering vector:
      V1      V2      V3      V4      V5      V6      V7      V8      V9     V10     V11
1  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
2  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66
1  2  1  2  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132
1  2  1  2  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165
1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198
1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231
1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264
2  2  2  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297
2  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330
2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363
1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396
1  2  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429
1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462
1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495
1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528
2  1  2  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561
2  2  1  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594
1  1  1  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
595 596 597 598
1  1  1  1
```

Figure 10: k-means Output

Step 2&3: Plot clusters & Choose k (plot total sum of squares)

Process:

Each observation is assigned to a cluster pre-determined in the code with k=2 based on the output of complete linkage technique (Fig.9). We use that information to generate Fig.10. It should be added that there is linear separability of the clusters in this method. Therefore, we will draw a line parallel to the y-axis(x=constant) to separate both clusters. We will discuss it further in the interpretation below.

Interpretation:

K=2 is proved to be the right approach as confirmed by both the cluster plot and WSS plot (Fig.10 & Fig.11). k-means technique is more appropriate for a data sets with a big dimension indeed.

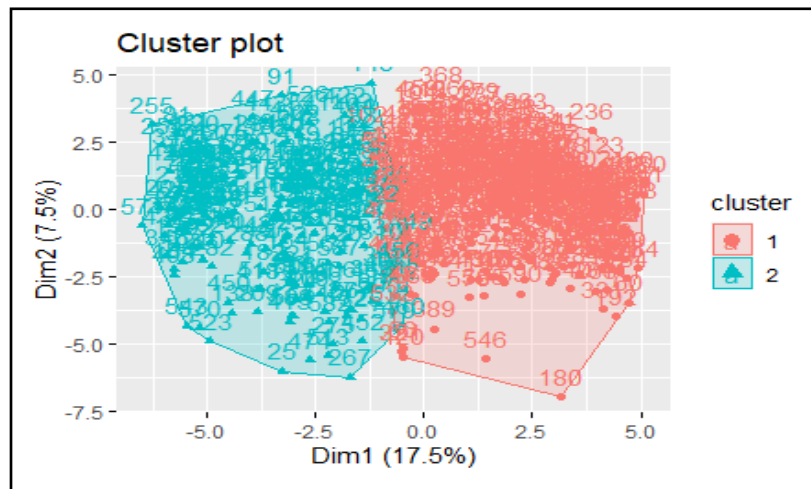


Figure 10: Cluster Plot (k-means technique). Cluster 1 is more homogeneous than cluster 2

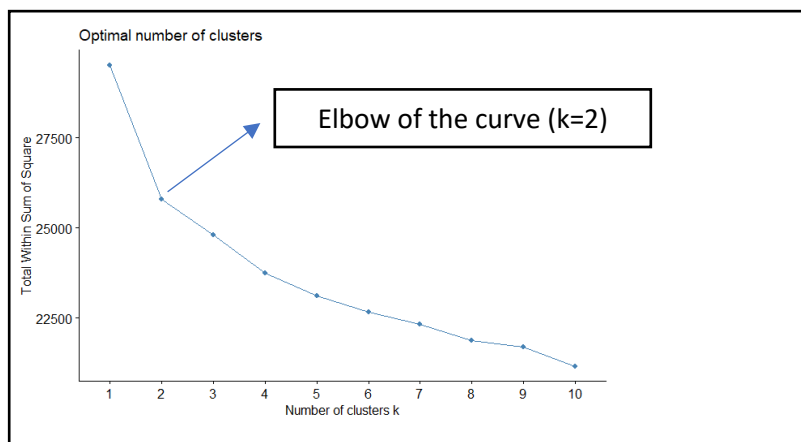


Figure 11: Plot Total Sum of squares. There is diminishing return by adding additional clusters over 2

Step 4: Interpret clusters

Interpretation:

Even though the results were skewed by variable 12 to 18 in cluster 2, we had a good response for the k means analysis. Regarding this cluster 2, we are looking for homogeneity which is difficult with high variance. Even though those variables are relevant, there are limiting the quality of our k-means clustering analysis. As a result, cluster 1 is less heterogeneous than cluster 2. Therefore, a set of clusters like cluster 1 would be ideal to have a quality answer to study the impact on this political decision on mass transportation demand for most people in the 3 southern cities. In my opinion, k means clustering is more adapted for this data than both single and complete linkage as the data is linearly separated using the former (**Fig.12**).

Furthermore, cluster 1 has a more positive response to the survey. It can be deduced that their political views are in line with the decision to support Israel by the US government. This is plausible especially for Americans living in the South in 1973-1974. At that time, it can be hypothesized that most Americans agree on foreign policies with President Nixon (Republican) in these 3 cities. On the other hand, economic concerns drove the negative response from cluster 2. To conclude each cluster represented the degree in which the economic strain by the oil embargo weighed more than their political views.

and/or the transportation costs relative to the responders 'income. In other words, Clusters 1 respondents are not strained financially enough to care or simply unfollow their political views while clusters 2 respondent can not afford to do so.

Step 5: Plot clusters with PCA

Process:

In a 3D plot, there are more principal components than PC1 and PC2 as shown in **Fig.12**. By plotting one more component, R provides the component with the third most variation to optimize the analysis of the plot. Compared to FOODP.DAT, a 2D screenshot is irrelevant due to the high dimension of the data.

Interpretation:

There is higher PC1 value for respondent in cluster 1 -on the right side of the separation line- than cluster 2 respondents on the left side of the line. Most of the respondents in cluster 1 are also above $y(PC2=-4)$ which represents the principal component with the second most variation. On the other hand, cluster 2 respondents dots are heterogeneous with lower values in principal component 1 and scattered dots along principal component 2. Even though respondents are unreliable when it comes to openly give their salary, this plot confirms the assessment made in step 4 which is that respondent's financial means was the driving force in clustering them. Strong response with high variance on the left side of the plot against moderate response on the right side of the plot. As a conclusion it would have been interesting to add a question about knowledge on middle east politics to the participants. As much as the clustering analysis highlight the social status of the respondents (financial standpoint), there is no indication to analyze the importance of their political beliefs as they answered the survey.

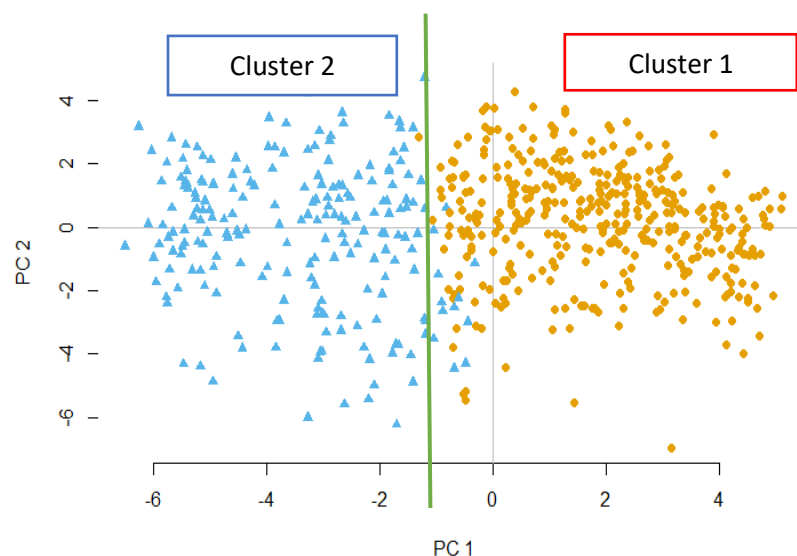


Figure 12: PC1 vs PC2 plot (Line Equation: $X(PC1) = -1$)

APPENDIX

• **FOODP.DAT**

Atlanta	Anchorage	Atlanta	Baltimore	Boston	Buffalo	chicago	Cincinnati	Cleveland	Dallas	Detroit
Baltimore	44.757011									
Boston	52.787877	14.944899								
Buffalo	41.650690	16.630394	22.376327							
Chicago	65.452425	33.485967	40.970477	30.240701						
Cincinnati	45.392290	12.084287	22.718715	27.736258	40.521846					
Cleveland	43.580156	12.924396	18.627668	12.212698	31.797012	22.843161				
Dallas	49.497778	13.987852	24.975788	20.461427	24.710929	18.690372	15.413630			
Detroit	43.051829	11.464292	24.336393	23.518503	36.044140	10.816654	18.355108	17.161294		
Honolulu	49.476257	15.683431	19.794444	11.033585	26.454489	26.310454	15.916344	17.088885	25.343638	
Houston	21.335885	29.112025	37.928617	30.737274	52.997358	29.277637	28.007320	34.601012	24.855181	38.416663
Kansas City	51.741086	16.776472	24.908232	29.978993	36.660878	13.519615	21.943336	13.459941	17.868968	26.410225
Los Angeles	56.737113	25.578702	23.381617	30.564031	41.250697	29.913876	22.750604	23.227355	32.879781	26.114747
Milwaukee	58.641282	22.929239	31.968735	32.506307	28.516311	21.974986	27.916841	13.216278	24.899398	26.101724
Minneapolis	57.614668	23.286477	34.330307	30.141168	25.145377	24.634326	30.796428	22.005908	21.626604	27.138718
New York	51.005784	14.420125	23.782136	16.469062	22.007499	23.028461	15.556992	14.658104	16.543579	16.558684
Philadelphia	34.462443	22.037922	28.870400	15.459625	38.492856	29.929918	14.112052	25.956695	22.522877	25.297826
Pittsburgh	31.734681	20.125109	23.660304	18.404619	46.060938	26.609961	16.819631	29.131770	22.613049	26.414201
St. Louis	47.900939	12.752647	20.649939	9.641577	23.623294	23.741946	11.236992	13.936642	19.086121	9.729337
San Diego	52.798580	18.949406	28.929051	21.715202	19.225764	24.578242	17.364907	6.697761	20.990236	18.911108
San Francisco	68.721030	30.998226	33.515668	39.964234	36.079634	31.105948	35.277046	24.259637	36.436520	30.998387
Seattle	50.219020	17.655028	30.497705	27.847980	29.253889	15.575943	23.771411	10.282023	16.380781	24.518768
washington	54.508990	12.821856	22.784863	23.876139	25.872572	16.029660	20.604368	11.074295	15.757855	18.898413
	45.912919	16.088508	22.533444	25.905137	39.403301	17.756354	14.793042	18.368236	14.320827	27.757055
		Honolulu	Houston	Kansas City	Los Angeles	Milwaukee	Minneapolis	New York	Philadelphia	Pittsburgh
Atlanta										
Baltimore										
Boston										
Buffalo										
Chicago										
Cincinnati										
Cleveland										
Dallas										
Detroit										
Honolulu										
Houston		35.378807								
Kansas City		43.213655	20.992856							
Los Angeles		44.401577	13.662357	26.761540						
Milwaukee		43.424532	28.225698	42.971968	23.553556					
Minneapolis		35.328317	24.372320	32.600767	24.670428	17.206394				
New York		20.162837	32.668333	35.303824	38.909382	36.481091	22.292824			
Philadelphia		17.798034	31.968891	33.748333	40.727509	39.616158	27.034423	12.170045		
Pittsburgh		33.990881	24.491223	28.488594	25.464878	22.582958	7.808329	19.527417	23.410041	
St. Louis		37.945619	18.830560	26.392234	15.786387	22.189637	13.952778	26.884010	32.436708	14.401042
San Diego		55.185687	21.550870	23.999583	15.295097	35.141002	34.640583	48.501753	48.461634	33.725658
San Francisco		35.566136	12.708658	29.947454	10.434079	18.810635	19.898241	32.007655	34.047614	21.527889
Seattle		38.732803	15.657905	28.643848	14.376022	14.641038	12.919752	30.825801	32.195652	15.538018
washington		26.420221	15.680883	24.068446	26.750852	32.564123	22.076777	22.027304	21.972030	22.346322
		St. Louis	San Diego	San Francisco	Seattle					
Atlanta										
Baltimore										
Boston										
Buffalo										
Chicago										
Cincinnati										
Cleveland										
Dallas										
Detroit										
Honolulu										
Houston										
Kansas City										
Los Angeles										
Milwaukee										
Minneapolis										
New York										
Philadelphia										
Pittsburgh										
St. Louis										
San Diego		26.615597								
San Francisco		14.216891	24.922881							
Seattle		14.040655	24.282710	11.777096						
washington		21.668551	34.900116	22.025079	21.660335					

Table 3: Unstandardized table highlighting the Euclidean distance between two vectors.

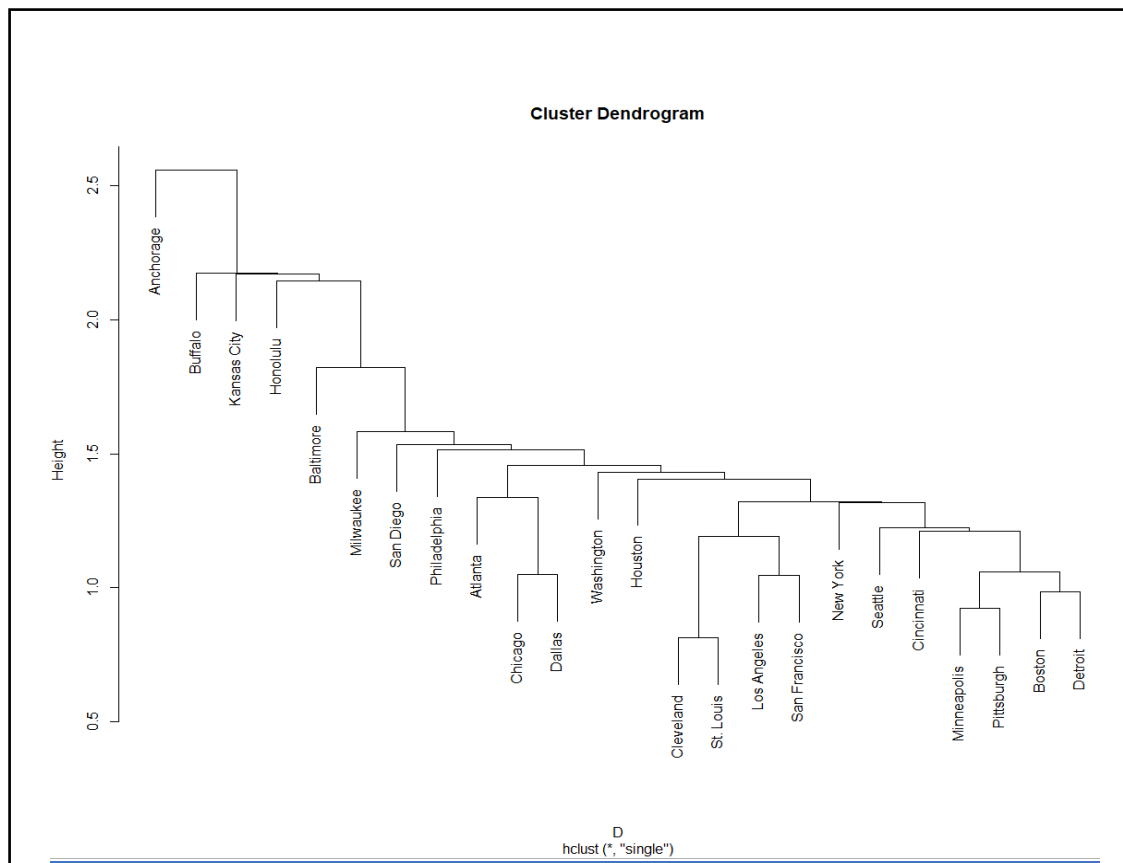


Figure 2: Standardized cluster dendrogram

```
> cl$centers
      Bread Hamburger      Butter      Apples      Tomato
1 -0.0262288  0.2787768  0.3947850 -0.6903737  0.7414787
2 -0.4263011 -0.1221365  0.4378053  1.1200502 -0.1024930
3  2.6843023  1.8399271  1.1365779  1.5208159  0.9787519
4 -0.3808157 -0.6333666 -0.8419380 -0.1453855 -0.9134267
```

Table 8: Cluster Center(standardized)

```
> cl$cluster
      Anchorage      Atlanta      Baltimore      Boston      Buffalo      Chicago      Cincinnati      Cleveland
      3          1          1          1          4          2          1          4
      Dallas      Detroit      Honolulu      Houston      Kansas City      Los Angeles      Milwaukee      Minneapolis
      2          1          3          2          1          4          4          4
      New York      Philadelphia      Pittsburgh      St. Louis      San Diego      San Francisco      Seattle      Washington
      1          1          1          4          4          4          4          2
```

Table 9: Standardized clusters (k-means)

- MASST**

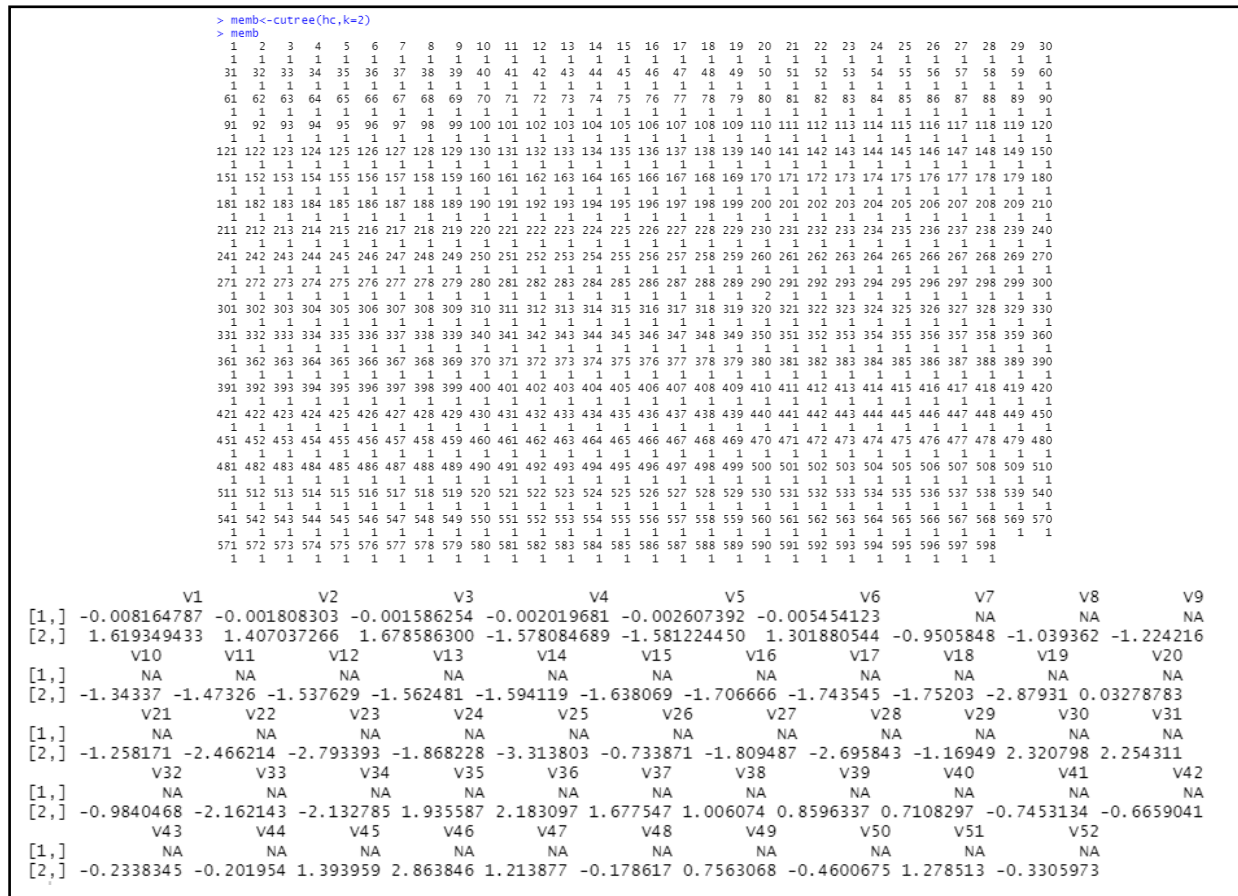


Figure 9B: Cluster Centers-single linkage- minimum distance approach do not work with the datasets

The following is a brief description of the data in file MASST.DAT
(Questions 4.9, 5.12, 6.9, 7.5, 8.9)

Variable	Variable Location (Columns)	Description
ID	1-4	Identification number from 1 to 598
V1	5-6	Economy
V2	7-8	Convenience
V3	9-10	Flexibility
V4	11-12	Safe from dangerous people
V5	13-14	Low energy use per passenger
V6	15-16	Dependability
V7	17	10% increase
V8	18	20% increase
V9	19	30% increase
V10	20	40% increase
V11	21	50% increase
V12	22	60% increase
V13	23	70% increase
V14	24	80% increase
V15	25	90% increase
V16	26	100% increase
V17	27	150% increase
V18	28	> 150% increase
V19-V38	29-48	Energy crisis opinions
V39	49	Miles to work or school
V40	50	Age
V41	51	Gender
V42	52	Marital Status
V43	53	Driver's License
V44	54	Number of adults in household
V45	55	Number of children in household
V46	56	Number of cars in household
V47	57	Respondent's education
V48	58	Spouse's education
V49	59	Years of residence in city
V50	60	Number of full-time workers
V51	61	Annual household income
V52	62	Ethnic background

Table 12: Description of all 52 variables for Mass Transit

References

- [Agglomerative Clustering \(machinelearninggeek.com\)](http://machinelearninggeek.com)
- [The Why, When and How of 3D PCA - BioTuring's Blog](#)
- <https://npl.stanford.edu/IR-book/completelink.html>